CLVer research note / 09 Nov 2026

Can six blood draws identify insulin preservation?

An exact inverse problem and a replay of paired mixed-meal tests from children and teens with recently diagnosed type 1 diabetes.

The result in plain English

C-peptide is a marker of insulin made by the body. After a meal, the concentration in a blood draw reflects both how much C-peptide was released and how much time the body had to clear it. A large release earlier between draws can leave the same reading as a smaller release closer to the draw.

Among **98 children** with two complete fasting tests, **66** had a lower C-peptide area-under-the-curve score at week 52. For **21 of those children**, there are two continuous secretion histories that fit every draw and give opposite directions of change. The histories use the same clearance model and satisfy explicit rate and slope limits.

What the paired samples force under each model

Admissible secretion	Decline	Increase	Either sign
Any nonnegative input	11	1	86
Continuous input with limits	45	8	45

Each row accounts for all 98 children. A forced direction holds for every history in that row's class. Either sign means both a decrease and an increase have explicit feasible histories.

The cohort mean can change in either direction

The mean concentration score fell from **0.632 to 0.530 nmol/L**. With the continuous-rate limits, the same individual samples support mean secretion changes ranging from a **2.83 nmol/L fall** to a **0.97 nmol/L rise**, to the shown precision. Secretion totals here are normalized by central distribution volume. The trial's concentration endpoint and the reconstructed secretion total are distinct quantities.

Scope. Paired complete-case analysis under common nominal kinetics. The conclusions concern assay interpretation and secretion per unit of central volume. Treatment effects and beta-cell mass require separate evidence.

A sharp set for secretion totals

Let C be the sampled central concentration and S the cumulative secretion per unit central volume over the test. The two-compartment impulse response uses positive weights and two clearance modes.

$$g(t)=ae^{-\alpha t}+(1-a)e^{-\beta t},\quad 0<a<1,\quad \alpha>\beta>0.$$

Write b_i for the concentration at the i-th draw after subtracting the propagated initial state. In sampling gap j of length Δ_j, use elapsed-time coordinate $x = \exp(-\beta(t_j - t))$. It ranges from $\ell_j = \exp(-\beta\Delta_j)$ to 1. Let $p = \alpha/\beta$ and retain three moments in each gap.

$$M_j=\int dS,\qquad u_j=\int x\,dS,\qquad v_j=\int x^p\,dS.$$

Theorem

The sharp possible totals are the values of ΣM_j subject to the sample equations and the following gap inequalities. If $M_j = 0$, set $u_j = v_j = 0$.

$$b_i=\sum_{j\le i}\left[ae^{-\alpha(t_i-t_j)}v_j+(1-a)e^{-\beta(t_i-t_j)}u_j\right].$$

$$M_j\ge 0,\qquad \ell_j M_j\le u_j\le M_j.$$

$$\frac{u_j^p}{M_j^{p-1}}\le v_j\le \ell_j^p M_j+\frac{1-\ell_j^p}{1-\ell_j}(u_j-\ell_j M_j).$$

Proof

Convolution gives the sample equations. In a gap with positive mass, divide by that mass to obtain a probability distribution. Jensen's inequality for x^p gives the lower bound and the secant line joining ℓ_j to 1 gives the upper bound. The single atom at u_j/M_j attains the lower bound. The endpoint mixture with the same mean attains the upper bound. Convex mixtures of these two measures realize each intermediate value. Multiply by M_j and join the gaps.

Endpoints in this sharp closed model are ordered events. A pulse at a gap's left edge occurs just after the preceding draw and one at its right edge just before the next draw. Each gap mass is bounded by the draw at its end because $g(\Delta_j)$ is positive. The feasible set is compact and convex, so its total masses fill a closed interval whenever the sample equations are feasible.

The data certificates use strictly interior pulse locations, and the continuous witness analysis uses ordinary nonnegative rates with no point masses.

How the trial data were checked

CLVer randomized 113 children ages 7–17. We paired the randomization and 52-week visits, requiring one fasting test day with six complete numeric C-peptide results at each visit [1–3]. Each day was grouped by participant, visit and collection date. A visit with two complete days was excluded. The public release yielded 99 complete numeric pairs and 98 after excluding the one nonfasting pair. Including that pair still gives 45 continuous-rate reversals.

Kinetics and rate limits

We used common nominal kinetics close to the published reference two-compartment parameters [4, 5]. With r = 93/94 and λ = ln(94/93) per minute, the fast and slow half-lives are 4.985 and 32.404 minutes. We initialized both compartments at steady state for the first draw, C_0.

$$g(t)=\frac{19}{25}r^{13t}+\frac{6}{25}r^{2t}, \qquad C_{\mathrm{free}}(t)=C_0\left(\frac{19}{58}r^{13t}+\frac{39}{58}r^{2t}\right).$$

At steady state the input is kC, where k = 1 / ∫g = 325λ/58. For the continuous class, the rate is linear between 5-minute knots and starts at kC_0. With R = k times the largest of the six observed concentrations at that visit, all knot rates lie between 0 and 2R and adjacent rates differ by at most 0.6R. These are sensitivity constraints chosen for the analysis.

Certificates cover the continuous time axis

A linear program proposes pulse masses and dual weights. For a lower total bound, its rounded dual must have weighted impulse response at most 1 at every possible release time. The reverse inequality gives an upper bound. In a gap, writing x = r raised to the elapsed minutes, the dual response is $A\,x^{13} + B\,x^{2}$. We check both endpoints and any interior critical point exactly. At such a point $x^{11} = -2B/(13A)$, so comparison with 1 reduces to a rational power comparison. Positive pulse masses are reconstructed by solving the sample equations over rationals.

For the continuous class, an outward rational interval encloses ln(94/93). One knot strictly inside each gap corrects the sample equation in time order. The checker validates all rate bounds after these corrections. Rounded duals bound the extrema after charging their residuals over the known rate box.

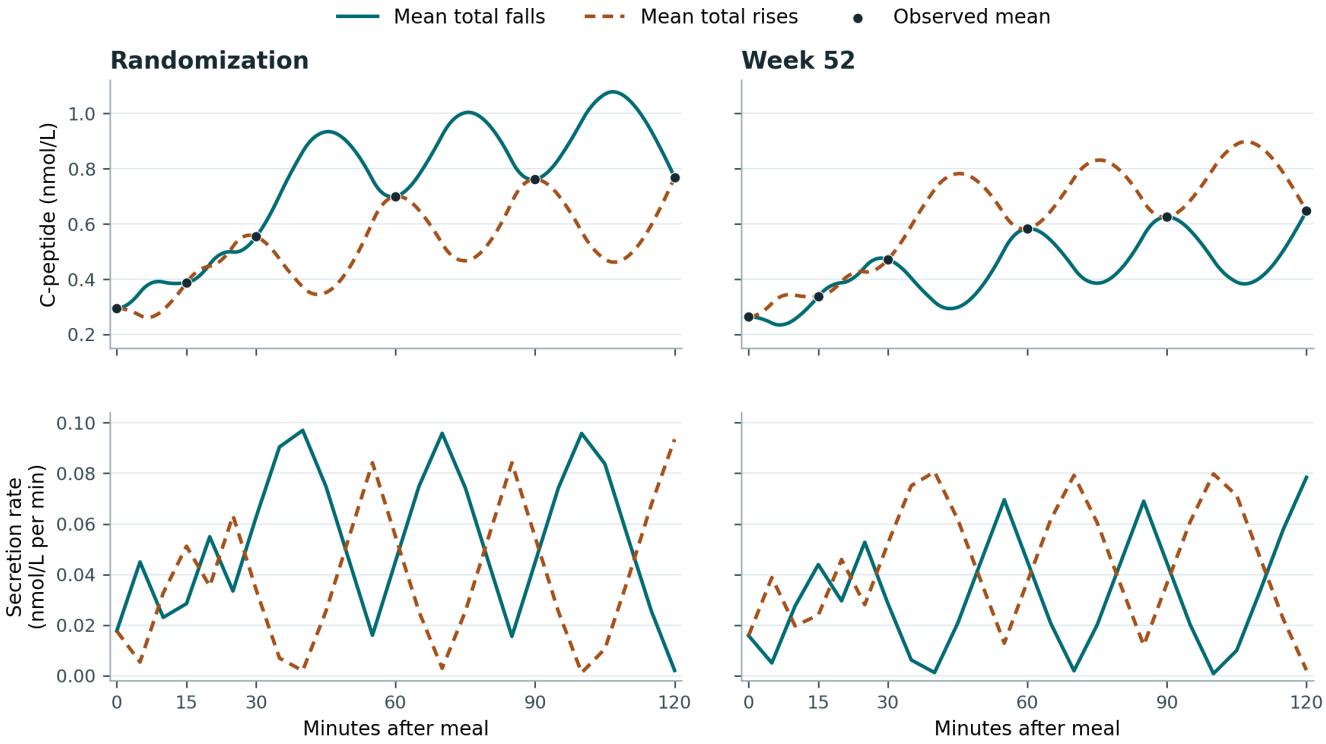
For the knot program, write Aq = d for the draw equations and initial rate, Gq ≤ h for the slope constraints, and 0 ≤ q ≤ Q for the rate box. For any equality weights y and slope weights μ ≤ 0, the objective $c^{\mathsf{T}}q$ has the following certified lower bound. Apply it to −c for the upper bound.

$$c^{T}q \;\geq\; y^{T}d+\mu^{T}h+Q\sum_{j}\min(0,c_j-(A^{T}y+G^{T}\mu)_j).$$

The largest inner-to-outer gap for a paired-change endpoint was below 1.5×10^{-7} nmol/L for the nonnegative class and 9.2×10^{-7} nmol/L for the continuous class. Exact comparisons classified all 98 fasting pairs.

Two histories that fit every draw

Each line below averages 98 feasible individual histories. Within a visit, both colored curves pass through the same six observed means. The lower panels show the secretion rates that produce those curves.



Compatible histories	At baseline	At week 52	Change
Mean total declines	6.551	3.726	-2.825
Mean total increases	4.482	5.450	+0.968

Values are secretion totals per unit central volume in nmol/L. Each comparison uses the same child at both visits, and each history fits that child's draws before aggregation. The plotted shapes exhibit the measurement ambiguity. A group mean trace is an illustration of the exact individual-level fits.

A concrete measurement that separates these witnesses

A draw at minute 105 would see a predicted separation of **0.597 nmol/L** at baseline and **0.498 nmol/L** at week 52 between these two sets of cohort histories. These are model predictions for the constructed witnesses. At both visits, the source tests report the six scheduled draws.

What one extra draw buys

Even if the complete two-compartment state at the start of the last gap were known, its final draw fixes only the contribution b from secretion in that gap. For a nonnegative input with total M and gap length Δ, monotonicity of g gives a sharp interval.

$$g(\Delta)M \le b \le M \qquad \Longleftrightarrow \qquad b \le M \le b/g(\Delta).$$

The endpoints come from putting all the input at opposite ends of the gap. For the nominal kinetics, the possible upper-to-lower ratio for positive b is **7.24** with a 30-minute gap, **3.72** with a 15-minute gap, and **1.68** with a 5-minute gap. This is a terminal-gap calculation conditional on the earlier state. Published comparisons of two- and four-hour tests address how much response lies outside the window [6]. The result here concerns timing ambiguity inside that window.

Scope and replay

The measured samples are treated as exact and placed at nominal minutes. Clearance and effective volume were not measured for each child. Missing, censored and repeated tests limit complete-case coverage. The reported ranges are model-based identification bounds. Statistical sampling error is outside these bounds. The source bundle replays the rational checks from the public archive and contains aggregate outputs only.

References

[1] McVean et al. Tight glycemic control in newly diagnosed pediatric type 1 diabetes. JAMA (2023).

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[3] Jaeb Center for Health Research. CLVer public dataset. October 2026 release.

[4] Van Cauter et al. Standard kinetic parameters for C-peptide clearance. Diabetes (1992).

[5] Ormsbee et al. Variability in estimated modelled insulin secretion. JDST (2022).

[6] Boyle et al. Two- and four-hour tests capture different C-peptide responses. Diabetes Care (2016).

Data attribution

The source of the data is the CLVer: Hybrid Closed Loop Therapy and Verapamil for Beta Cell Preservation in New Onset Type 1 Diabetes: A Proof-of-Concept Study, but the analyses, content and conclusions presented herein are solely the responsibility of the authors and have not been reviewed or approved by the Jaeb Center for Health Research.